

Towhid Ahmed

AI/ML Engineer | GenAI Engineer

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PROFESSIONAL SUMMARY

Entry-level AI/ML Engineer with strong foundations in Machine Learning, Deep Learning, NLP, and Generative AI, along with solid Data Structures & Algorithms skills (400+ problems solved). Built and deployed end-to-end ML systems using Flask/FastAPI with model training, evaluation, and production deployment on cloud platforms like Render. Focused on developing scalable, real-world AI solutions with strong problem-solving ability.

CORE SKILLS

Programming: Python, Java, SQL

Machine Learning & Deep Learning: Supervised & Unsupervised Learning, EDA, Data Preprocessing, Feature Engineering, Regression, SVM, KNN, Decision Trees, Random Forest, Clustering, PCA, Gradient Descent, Neural Networks, CNN, RNN/LSTM. Model Evaluation (Accuracy, Precision, Recall, F1, Cross-validation)

NLP & Generative AI: Text Preprocessing (Tokenization, Stopwords, Lemmatization), Bag of Words, TF-IDF, Word Embeddings, BERT, Transformers, LLMs, Fine-tuning, RAG, Vector Databases (FAISS), LangChain / LlamaIndex, Prompt Engineering, AI Agents, Chatbots.

MLOps & Deployment: MLflow, DVC, DagsHub, Docker, Flask, FastAPI, REST APIs, CI/CD, Render/Railway

Software Engineering: Solve 400+ DSA Questions(Array,String,DP, Graph,Trees etc.), OOP, System Design basics.

Tools & Frameworks: TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, vscode, Git, GitHub,

EXPERIENCE (PROJECTS-BASED)

SMS Spam Detection System | Python, Scikit-learn, NLTK, Pandas, NumPy, Streamlit | Demo | GitHub

- Built an end-to-end NLP-based SMS Spam Detection system that classifies messages as Spam or Ham using machine learning and text preprocessing techniques.
- Model Performance: **Accuracy: 97.06% | Precision: 100%** | Best Model: Multinomial Naive Bayes

Production-Ready MLOps Pipeline (ETL + Model Lifecycle) | Python, Scikit-learn, MLflow, Docker, GitHub Actions | DagsHub

- Designed and implemented an end-to-end machine learning pipeline with modular architecture covering data ingestion, preprocessing, model training, and evaluation.
- Integrated MLflow for experiment tracking, parameter logging, and model versioning to ensure reproducibility and lifecycle management.
- Automated CI/CD workflows using GitHub Actions and containerized the application using Docker for consistent deployment across environments.

Potato Disease Classification | Python, TensorFlow, Keras, CNN, OpenCV, NumPy, Matplotlib, Pillow, Streamlit **Demo | GitHub**

- Developed a deep learning-based image classification system to detect potato plant diseases (Early Blight, Late Blight, Healthy) using Convolutional Neural Networks (CNN).
- Achieved high classification accuracy (~97.8%)
- Deployed interactive web app using Streamlit for real-time prediction

Movie Recommendation System (ML +NLP) | Python, Scikit-learn, Pandas, NumPy, TF-IDF, Cosine Similarity, Streamlit **Demo | GitHub**

- Built a content-based movie recommendation system using NLP techniques to analyze movie metadata (overview, genres, keywords, cast, crew) and generate personalized recommendations.
- Implemented TF-IDF vectorization and cosine similarity to compute content similarity and deliver real-time movie recommendations through a Streamlit interface.

Medical Chatbot (RAG-based LLM Application) | Python, LangChain, Flask, Pinecone, OpenAI GPT, Render, CI/CD **Demo | GitHub**

- Built a retrieval-augmented generation (RAG) medical chatbot using LangChain and OpenAI GPT to deliver context-aware question answering from external medical documents.
- Integrated Pinecone vector database for semantic search and deployed a Flask-based API for real-time LLM responses.
- Deployed on Render using Docker, with CI/CD automation via GitHub Actions for production-ready deployment.

EDUCATION AND CERTIFICATES

EDUCATION: Bachelor of Business Administration (BBA), Management, Nagirhat University, Bangladesh

CERTIFICATIONS: Mathematics - Linear Algebra, Probability, Statistics, Calculus – Udemy

- DSA, Machine Learning & Deep Learning – YouTube
- MLOps – Udemy
- NLP & Generative AI – Udemy